

ANKIT KUMAR

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EDUCATION

- **Birla Institute of Technology, Mesra** 2026
B.Tech in Electronics and Communication Engineering CGPA: 8.29
- **Narbheram Hansraj English School** 2022
Indian School Certificate (ISC), New Delhi Percentage: 95
- **Hill Top School** 2020
Indian Certificate of Secondary Education (ICSE), New Delhi Percentage: 94.2

EXPERIENCE

- **Samsung Research & Development, Noida** Jan 2026 - Jun 2026
SDE Intern Noida
 - Built a **Flask-ADB (Android Debug Bridge)** diagnostic utility adding **illegal IMEI detection** at the customer service tier, removing engineering-escalation dependency and cutting diagnosis time to under 60 seconds; in staged rollout to **300+ Samsung service centers** nationwide.
 - Rebuilt an internal file-transfer server (**Django, Nginx**) that hard-failed above 30 MB, implementing **chunked multipart uploads** with client-side segmentation, **async server-side reassembly**, and SQL-backed chunk-state tracking; removed the size cap, enabled **resumable uploads**, deployed across **5 teams serving 200+ users**.
 - Designed a Carrier Aggregation extraction engine using **PyQt6 and Pandas** indexing 500+ config entries across 10+ Qualcomm spec files into a unified searchable interface, cutting engineer lookup time by **95%+** (5+ min to **under 10 sec**).
- **Tata Motors** May 2025 - Jul 2025
SDE Intern Jamshedpur
 - Built a fault-isolated data pipeline replacing twice-weekly manual transfers with 15-minute aggregation cycles across **11 assembly-line sensor systems (165+ Excel reports per run)**; engineered per-source fault isolation for file locks, network drops, and machine-offline events, ensuring **zero-downtime** continuity.
 - Built a real-time safety-compliance system (**YOLOv10, Python**) processing **5 live CCTV feeds at 10 FPS** on standard laptop hardware, with automated **helmet-violation detection** and bounding-box audit logging, achieving **94%+ accuracy** on Bay 11 of the production line.

PROJECTS

- **Distributed Network Monitoring System**  Jun 2026
Tools & technologies :Java, Spring Boot, Apache Kafka, MongoDB, ShedLock, REST API, Docker
 - Built a horizontally scalable, **fault-tolerant** distributed system with independent scheduler, worker, and ingestion roles (one **Spring Boot** app via profiles) over partitioned **Apache Kafka** topics, scaling out as **consumer groups**.
 - Guaranteed single-active scheduling via a **MongoDB-backed distributed lock (ShedLock)** and **ordered, idempotent** processing through device-ID partition keying; containerized with **Docker**.
- **Predictive Maintenance (RUL Prediction)**  Feb 2025
Tools & technologies: Python, TensorFlow, XGBoost, Pandas, Random Forest, Scikit-learn
 - Built a **RUL prediction** system on **NASA C-MAPSS (FD001)**, removing 7 redundant sensors via Pearson correlation with RUL clipping and 10-cycle rolling-mean augmentation, achieving **RMSE 32.48** (R2 0.71) with **SVR** and 4-fold cross-validation.
 - Extended it to a **3-tier risk classifier** (No Risk, Moderate, Critical) using **SVC** and **Random Forest** with stratified splits, reaching **68%** validation accuracy across imbalanced classes.
- **Computed Tomography: Reconstruction of image using FFT**  August 2025
Tools & technologies : Python, Image Processing, FFT, FBP
 - Implemented a sinogram-based image reconstruction algorithm used in CT Scans.
 - Fine-tuned the algorithm's performance by reducing computational time through efficient use of FFT for filtering and interpolation, demonstrating the effectiveness of various reconstruction filters in enhancing image clarity.

TECHNICAL SKILLS

Languages: Java, Python, SQL, C++

Embedded/Systems: Linux, ADB, Shell, networking interface

Backend: Spring Boot, REST APIs, Microservices, Kafka, OpenFeign, Django, Flask

DevOps: Docker, Kubernetes, Git, GitHub Actions (CI/CD), Nginx

Databases: MySQL, MongoDB

AI / ML: LangChain, Groq API, SQLAlchemy, Scikit-learn, XGBoost, SVR, Random Forest, YOLOv10, Plotly/Dash

Tools & Concepts: Git, ADB, TCP/IP, HTTP/HTTPS, OSI Model

ACHIEVEMENTS

- **Leetcode Knight** (rating 1900+) Jun 2026
- **Graduate Aptitude Test in Engineering (GATE) in DA** Secured AIR 1388 Feb 2025
- **Joint Secretary, Robolution Robotics Club** Feb 2024